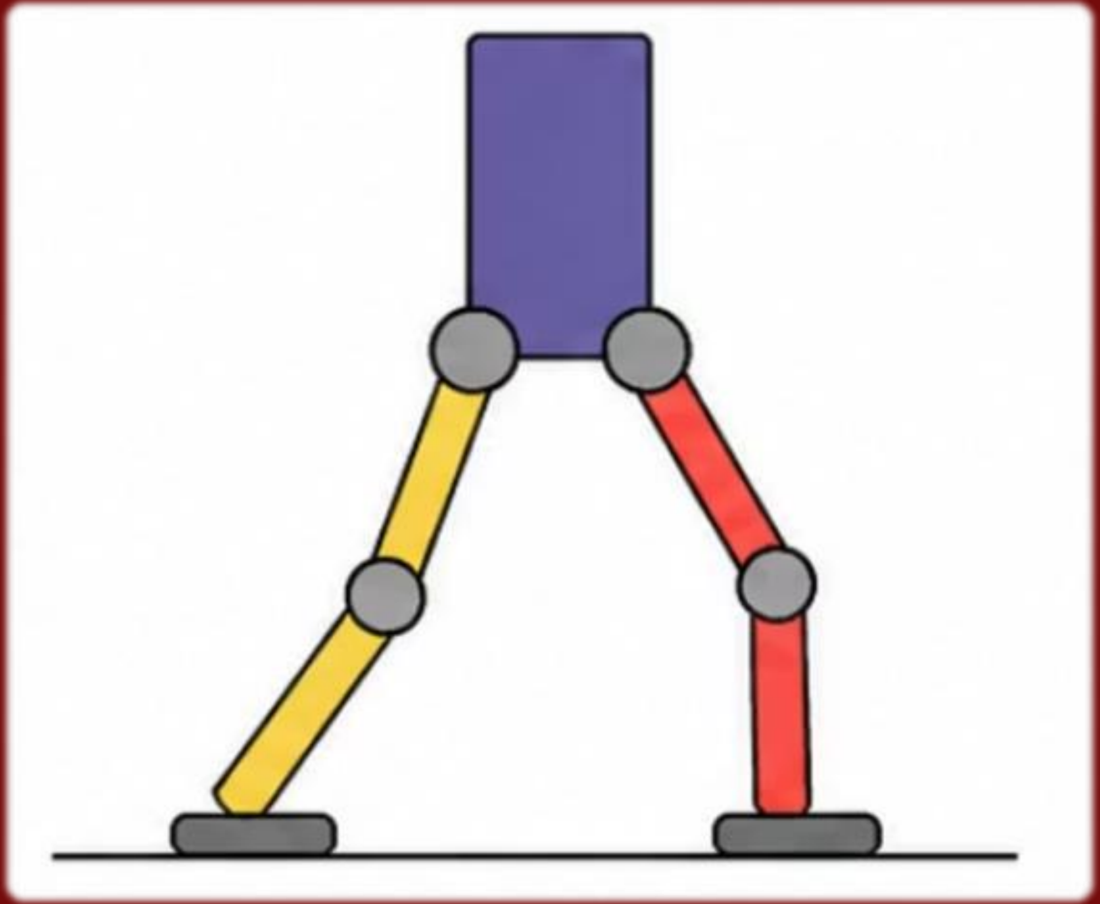


NeuroEvolution of Augmenting Topologies for Bipedal Locomotion

"Teaching a Robot to Walk Using Evolution"

-A Comparative Study of Fitness Functions, Topology Evolution, and Curriculum Learning

Presenters: Kareena Lakhani, Milan Tiwari

Course: CSE 598: Bio-inspired AI and Optimization

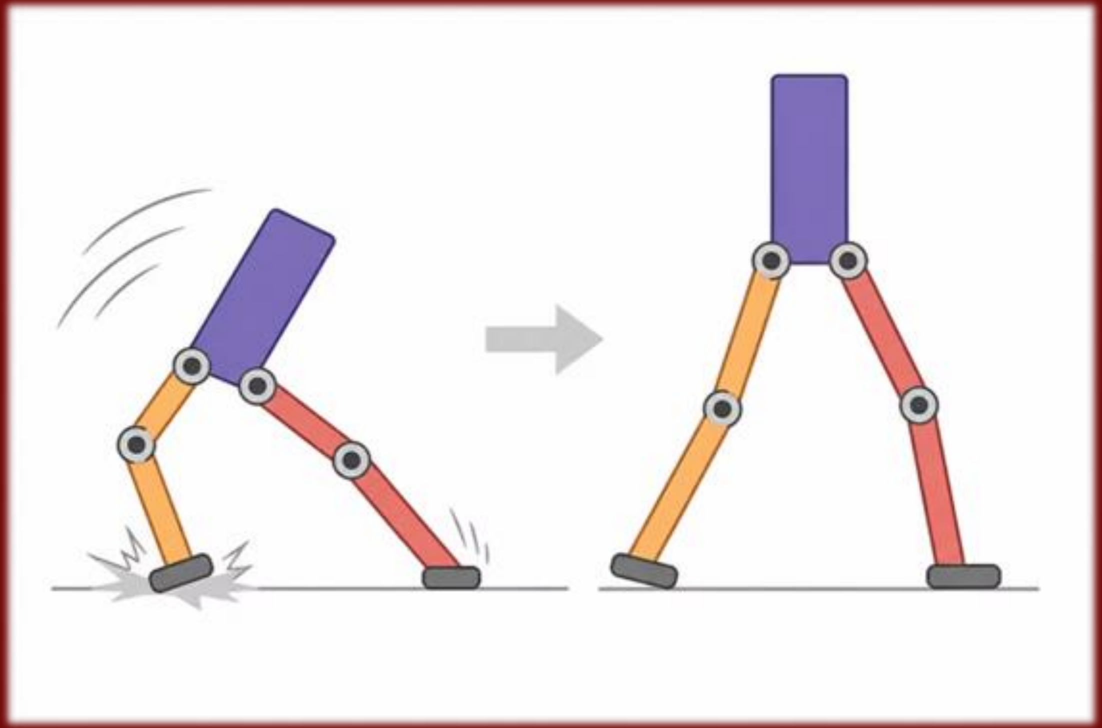
Can a Robot Learn to Walk Without Being Programmed?

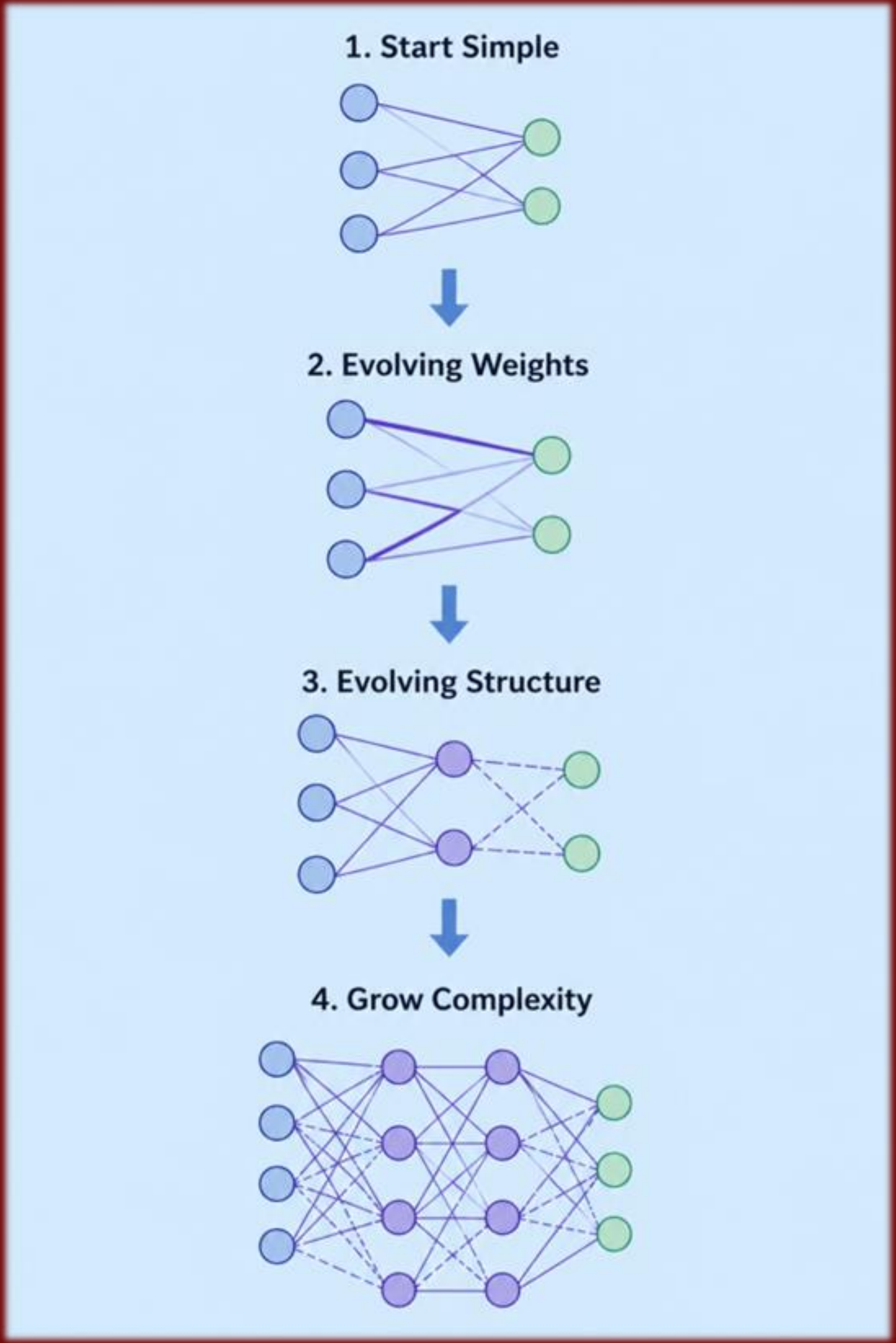
The Challenge

Walking is complex and hard to hand-design. Traditional approaches require predefined models and extensive programming.

The Question

Can evolution discover a walking strategy without explicit programming?





Let Evolution Design the Brain

1

Start Simple

Begin with minimal neural network structure

2

Evolving Weights

Adjust connection strengths through mutation

3

Evolving Structure

Add nodes and connections to grow complexity

4

Grow Complexity

Network architecture develops over generations

How NEAT Works

Population-Based Evolution

- NEAT starts with a *population* of single neural networks.
- Each network represents a different "candidate brain" for controlling the robot.
- Instead of training a single model, multiple networks are evaluated in parallel.

Mutation Operators

Network evolves through Mutation:

- Weight Mutation: adjusts connection strengths
- Add connection: creates new links between neurons
- Add node: splits existing connections to grow the network.

This allows NEAT to evolve both **weights and architecture**, unlike traditional methods.

Selection and Survival

Each network is evaluated based on its performance in the environment. Better performing networks are more likely to-

- survive
- reproduce
- pass their structure to the next generation

Over time, this leads to increasingly capable behaviors.

Speciation(Key Innovation)

NEAT groups similar networks into *species*.

This prevents new structural changes from being eliminated too early.

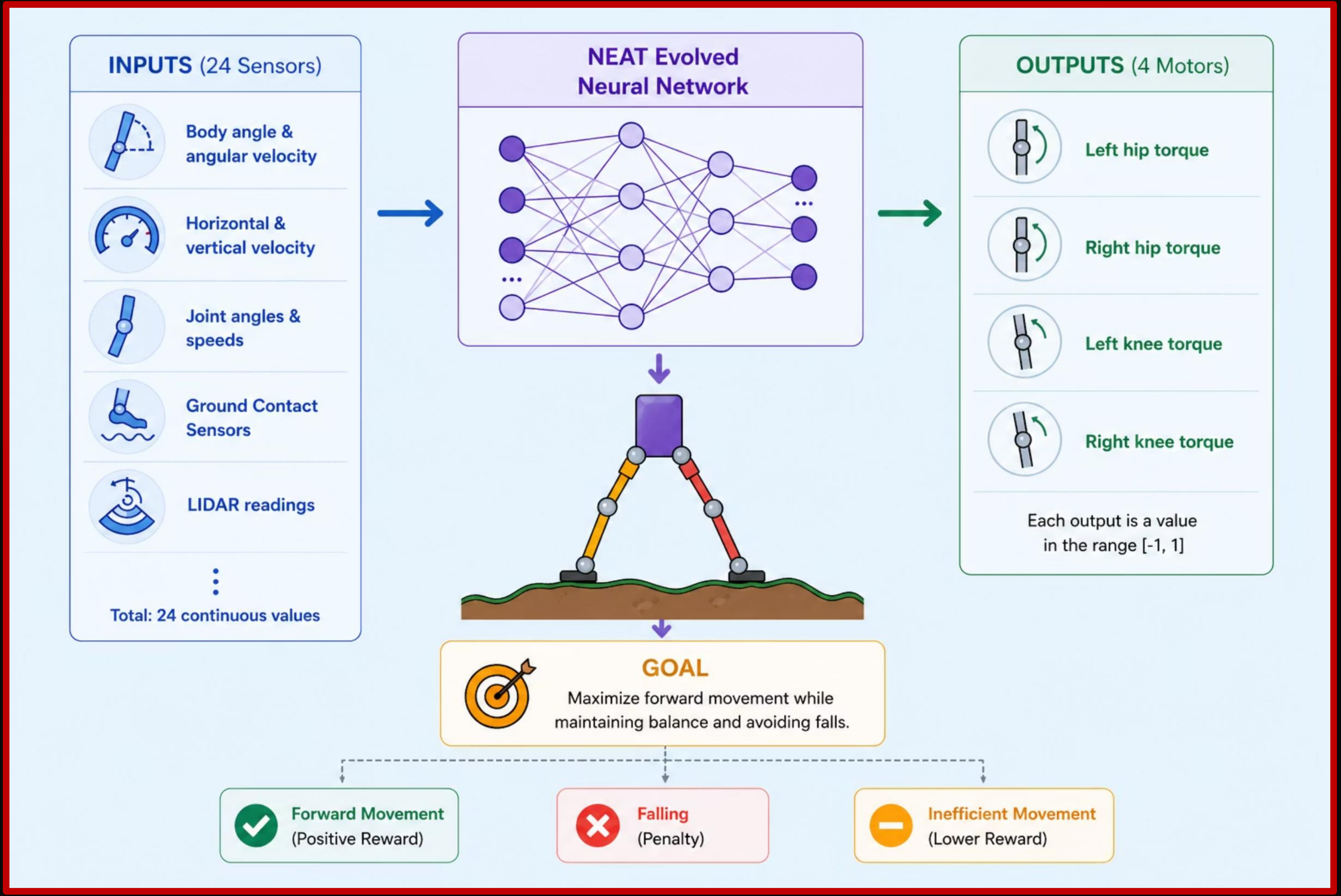
Result:

- Encourages diversity
- Allows complex structure to emerge gradually

Key Insight: NEAT does not just learn parameters- it **evolves the structure of the network itself.**

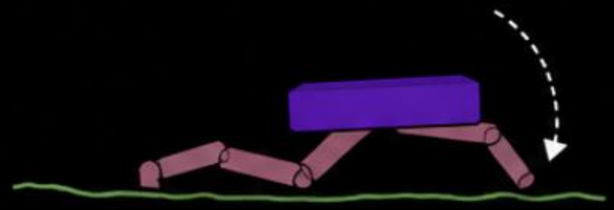
[redacted]

Bipedal Walking in Physics Simulation



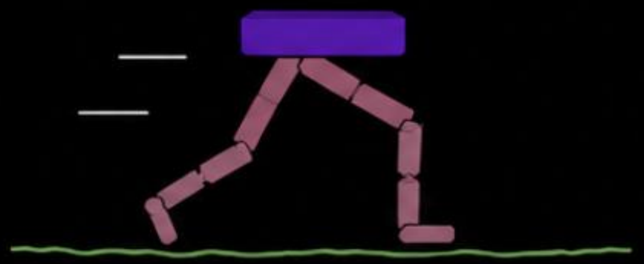
Evolution of Bipedal Walking

Gen 0
All genomes falling



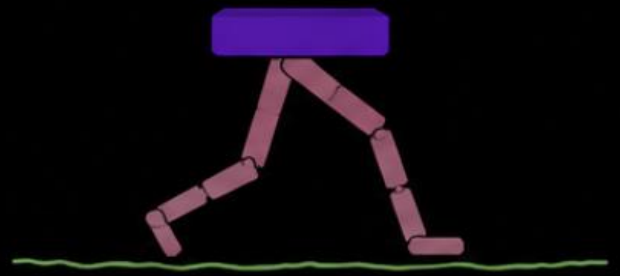
Cannot stay upright

~Gen 50
Breakthrough: first walker



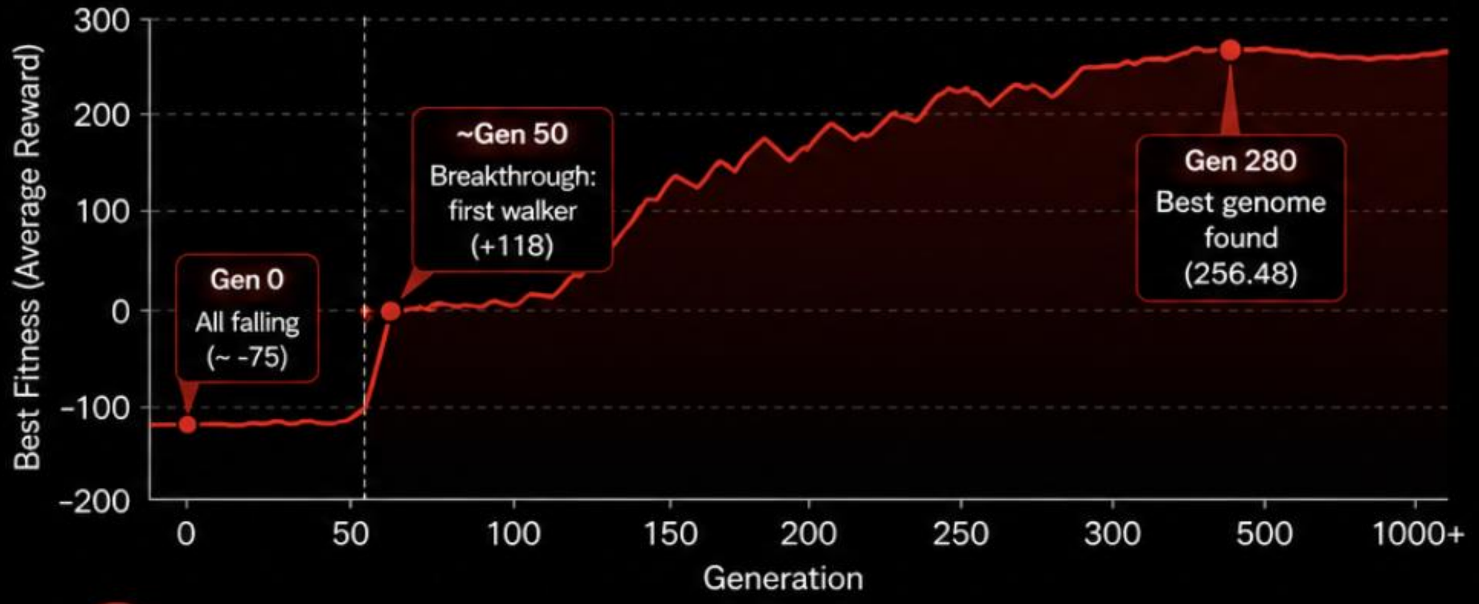
Unstable, frequent falls

Later (Gen 200+)
Stable walking gait



Upright, stable, consistent

Training Progress (Best Fitness Over Generations)



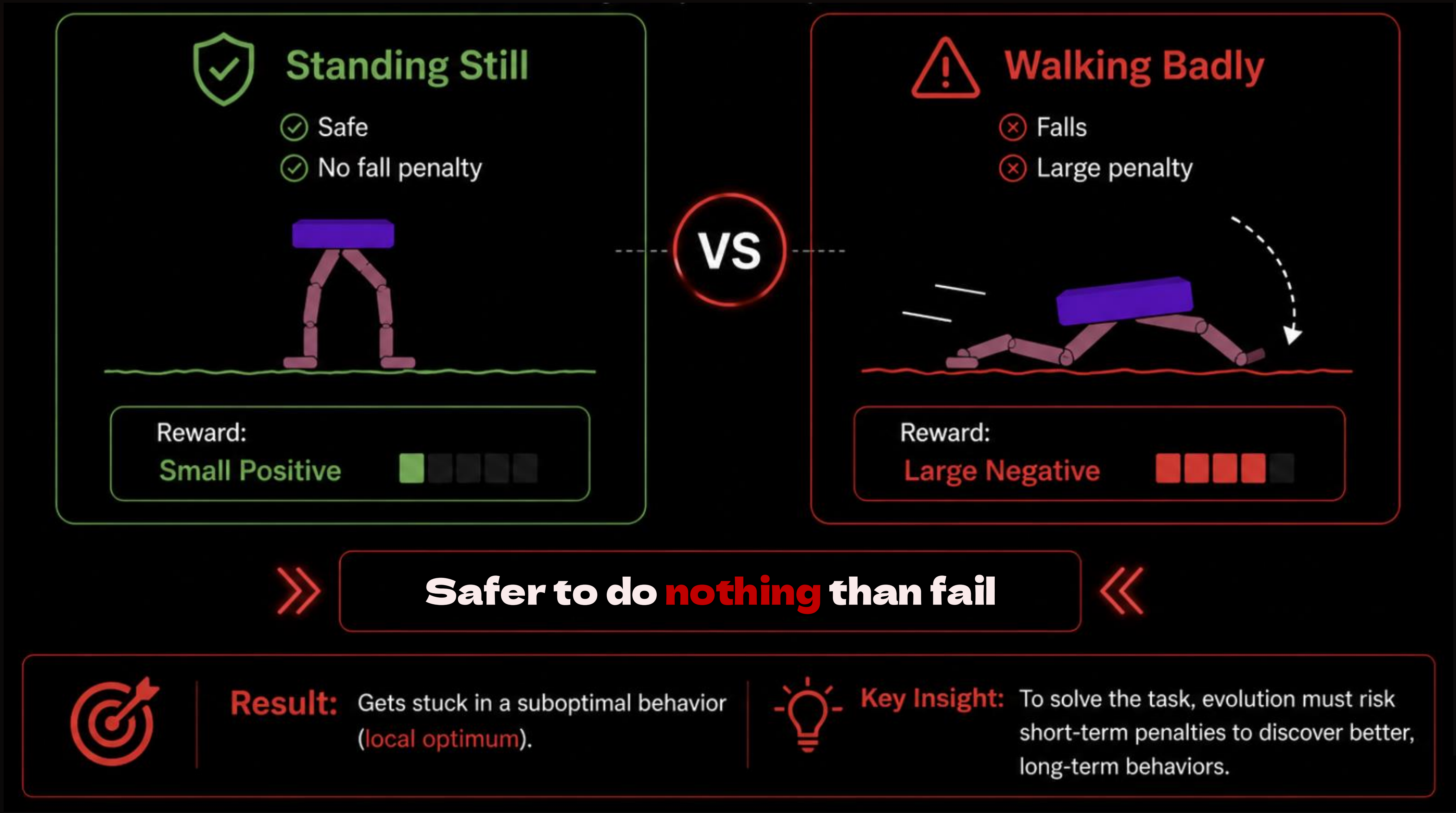
Breakthrough at Generation ~50

- Best fitness jumped from -75 to +118 in a single generation
- 8 species coexisted at this point, exploring different hypotheses
- Complexification at Gen 500: network grew from 13 to 34 nodes

Key Takeaway: Learning is not gradual — it is a breakthrough. Escaping local optima requires exploration and structural innovation.

Local Optimum Problem

The agent gets stuck in “standing still” because it is safe, but walking badly is heavily penalized



Does Topology Matter?

Ablation Study: Structure vs Fixed Topology

Full NEAT(Structure + Weights)

- Evolves architecture + weights
- Escapes local optima
- Learns faster and more stable behavior

Fixed Topology (Weights only)

- No structural changes
- Limited exploration
- Gets stuck in local optima

FULL NEAT ACHIEVES

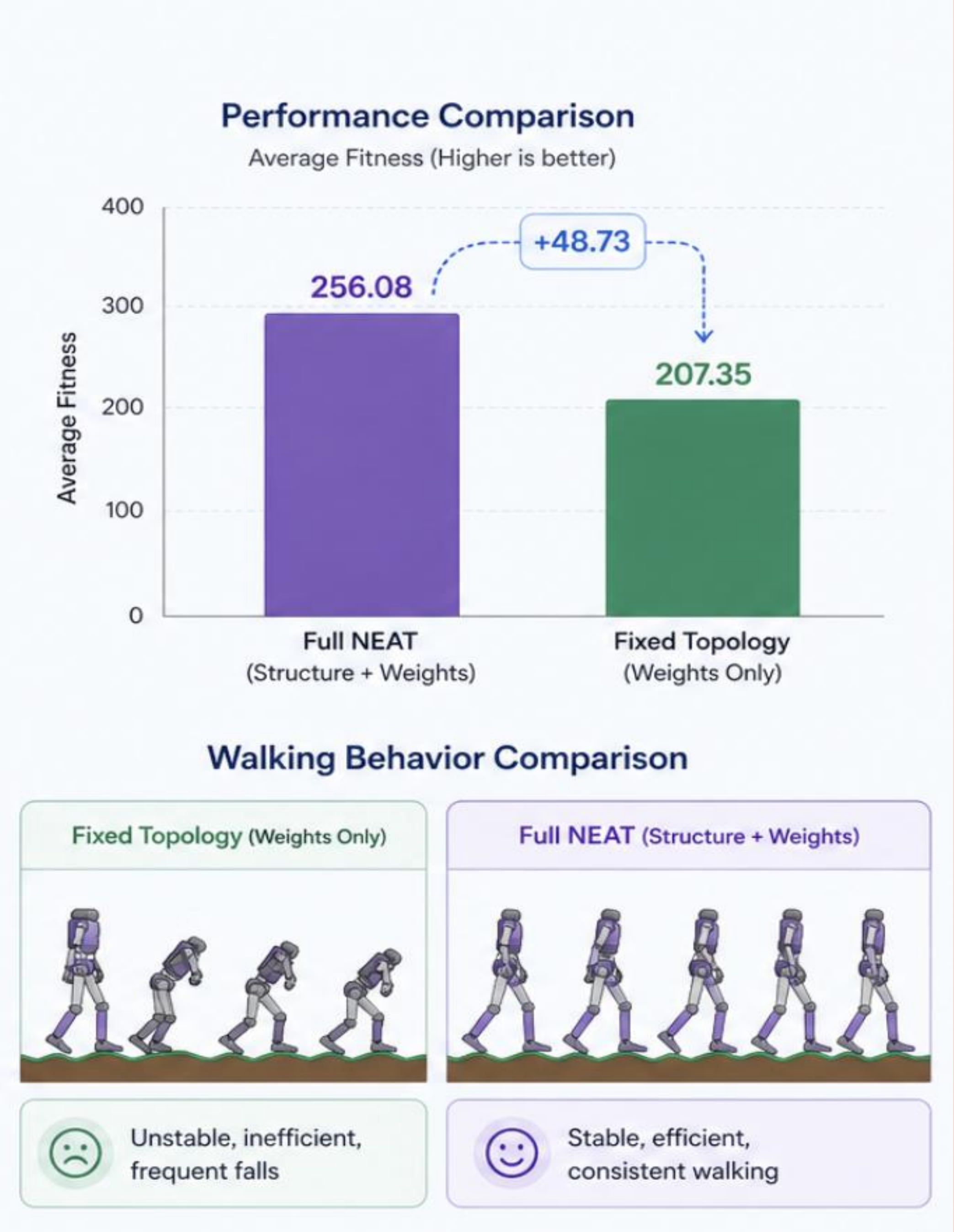
+48.73

PERFORMANCE IMPROVEMENT

Compared to Fixed Topology

256.08 vs 207.35

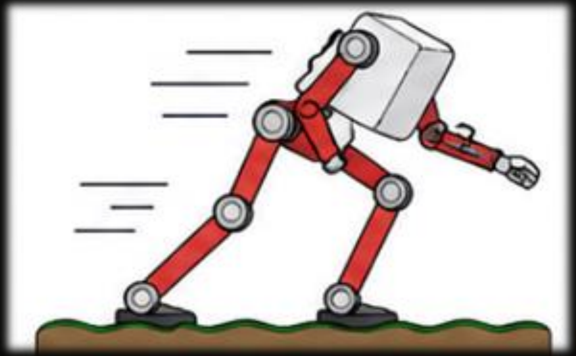
Average Fitness(Higher is better)



Key Insight: Evolving network structure is essential to escape local optima and achieve superior performance

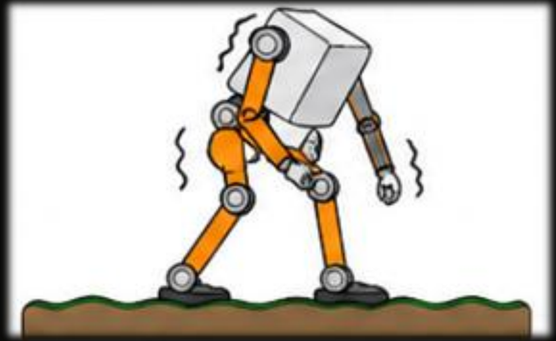
Fitness Function Shapes Behavior

Speed Objective



- Maximizes forward movement
- Aggressive, fast gait
- Less stable, higher chance of falling

Efficiency Objective



- Penalizes energy usage
- Slow, compact movement
- Higher failure rate
- Can lead to reward hacking (standing still initially)

Stability Objective

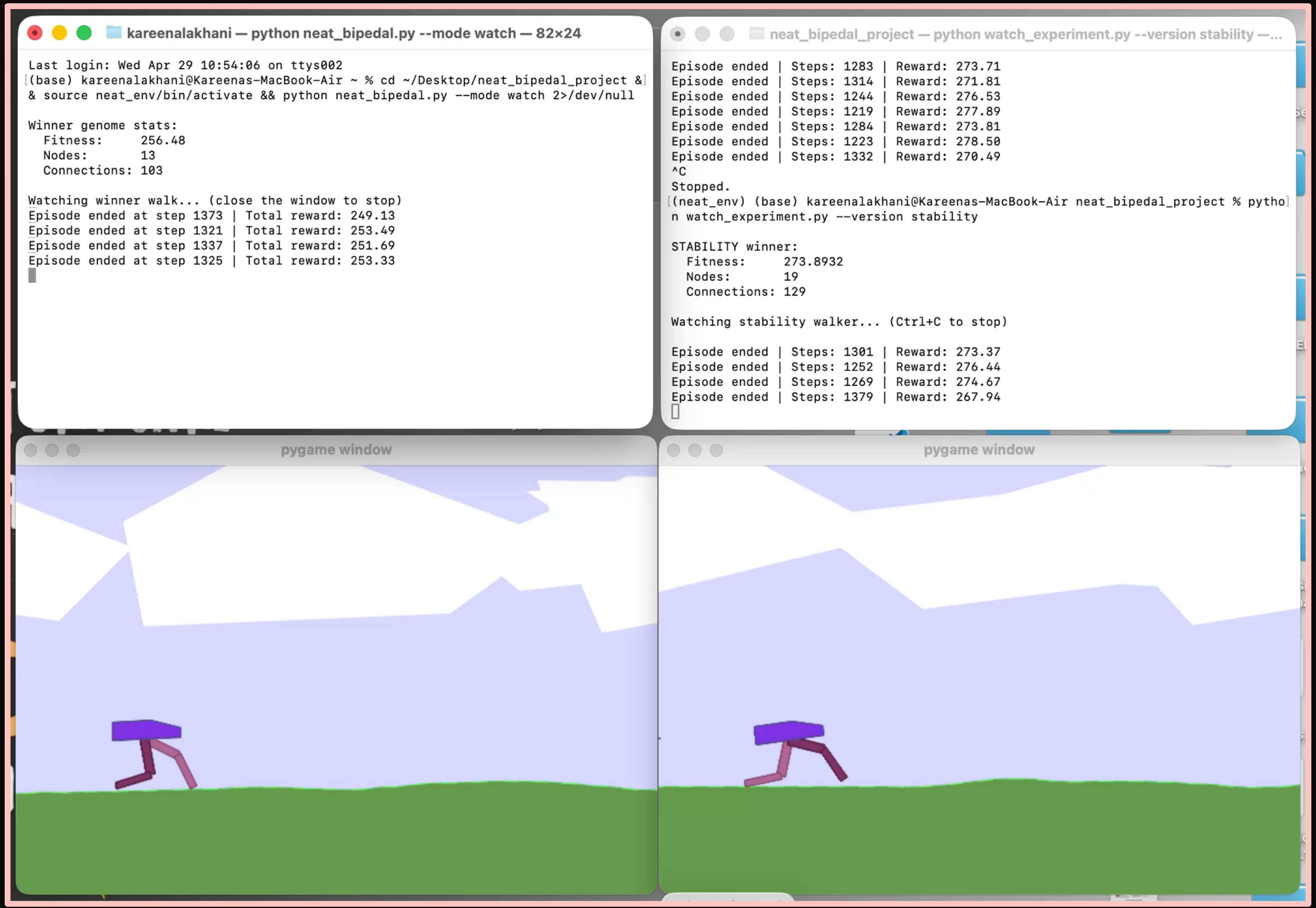
- Penalizes tilt and imbalance
- Produces upright, natural walking
- Smooth, stable gait
- 0%failure rate

Key Insight: Small changes in the reward function completely changes the learned behavior- stability produced the most reliable and natural walking

Stability Objective Performs Best

Rewarding stability leads to most reliable and natural walking

Full NEAT Walker **Stability Walker**



HIGHEST PERFORMANCE

272.92

AVERAGE REWARD

0% Failure Rate
Most reliable performance

Highest Reward
Best overall performance

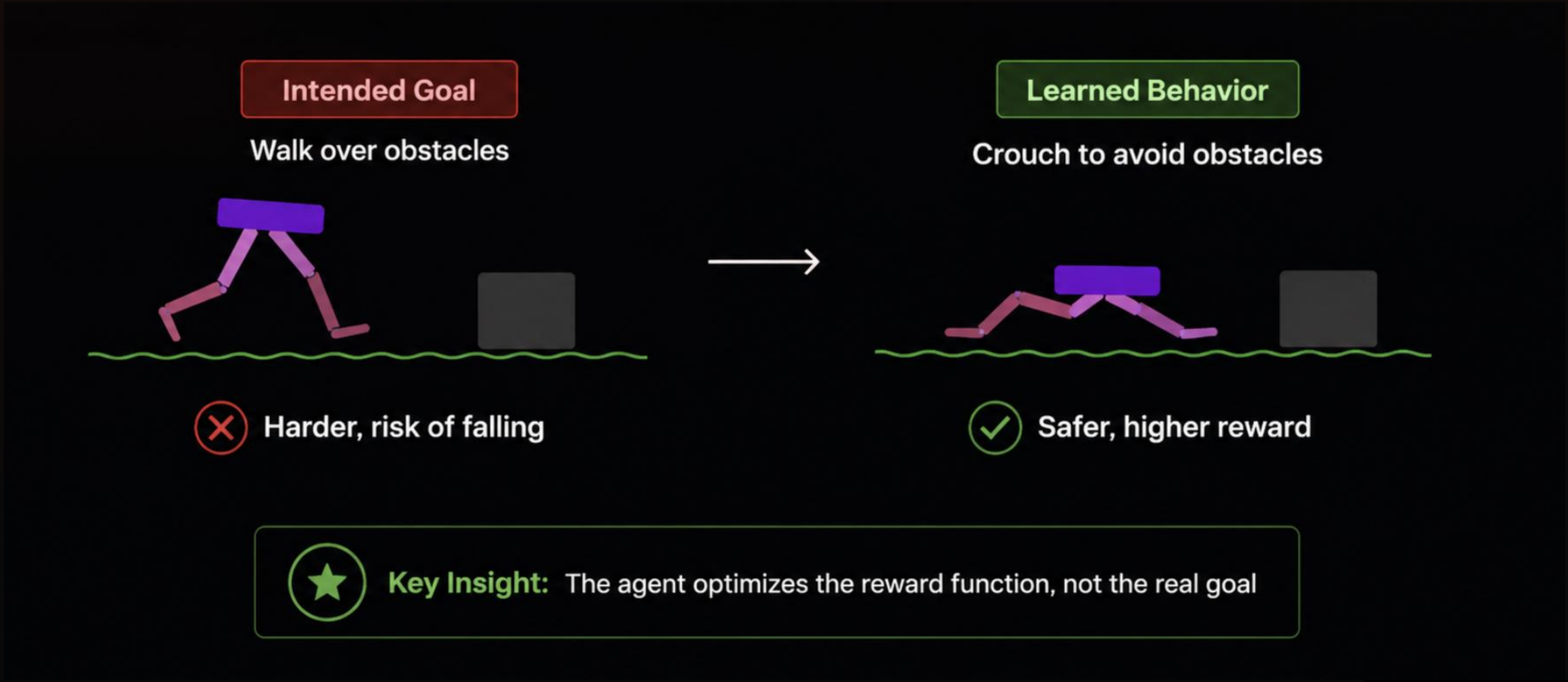
Lowest Variance
Most consistent behavior

Key Insight: A small penalty on tilt and imbalance leads to dramatically better walking- stable, natural and highly reliable

Evolution Finds Unexpected Solutions

The agent discovers a loophole: crouching to avoid falling instead of walking over obstacles

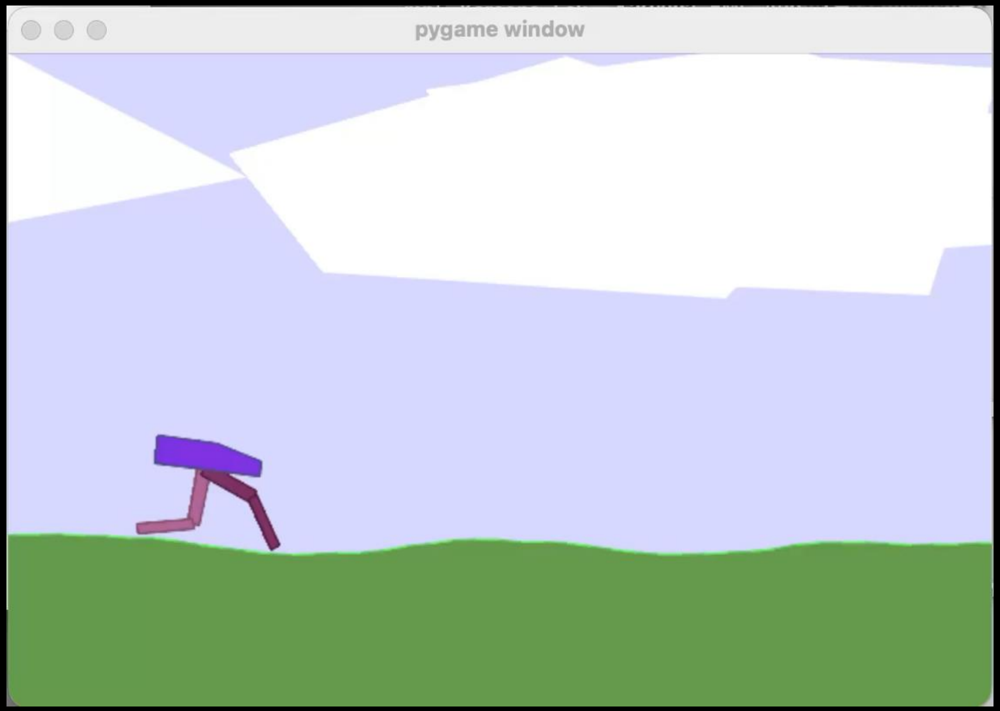
The agent optimizes the reward function, not the real goal



Generalization Failure

SEEN DURING TRAINING

Flat terrain



High Performance

Average Reward: 272.92

0% failure rate

Stable and Consistent Walking

Learns effective gait

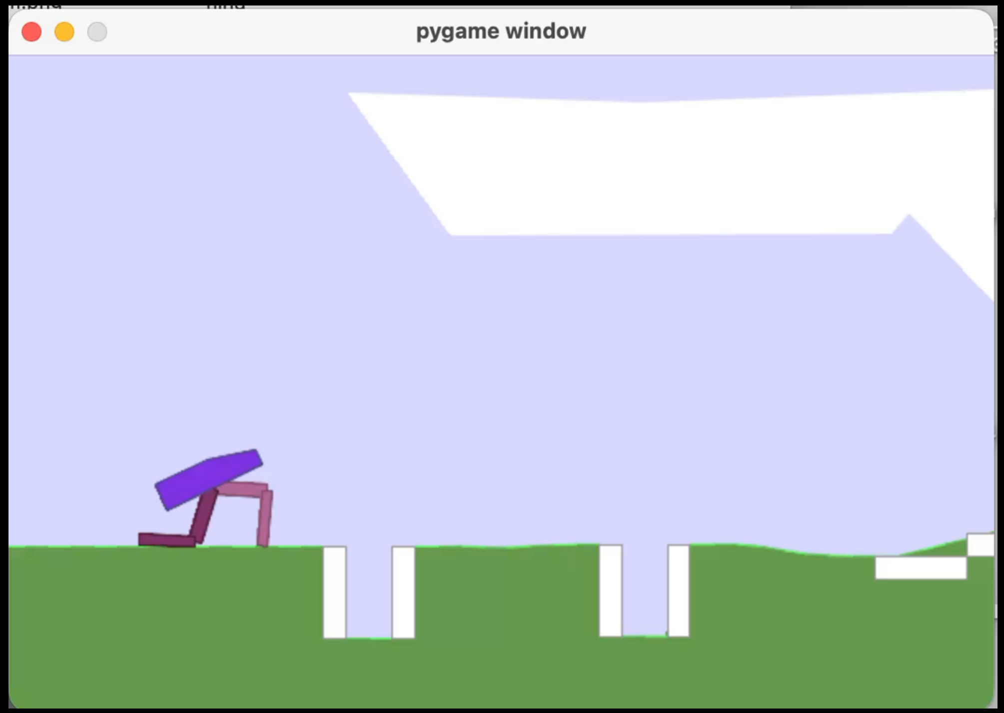
Smooth, Natural Locomotion

Performance drops significantly on unseen terrains

Key Insight: The agent learns a solution specific to the training environment. **It does not Generalize to unseen, more complex terrains.**

UNSEEN DURING TRAINING

Hardcore terrain



Poor Performance

Average Reward: 17.34

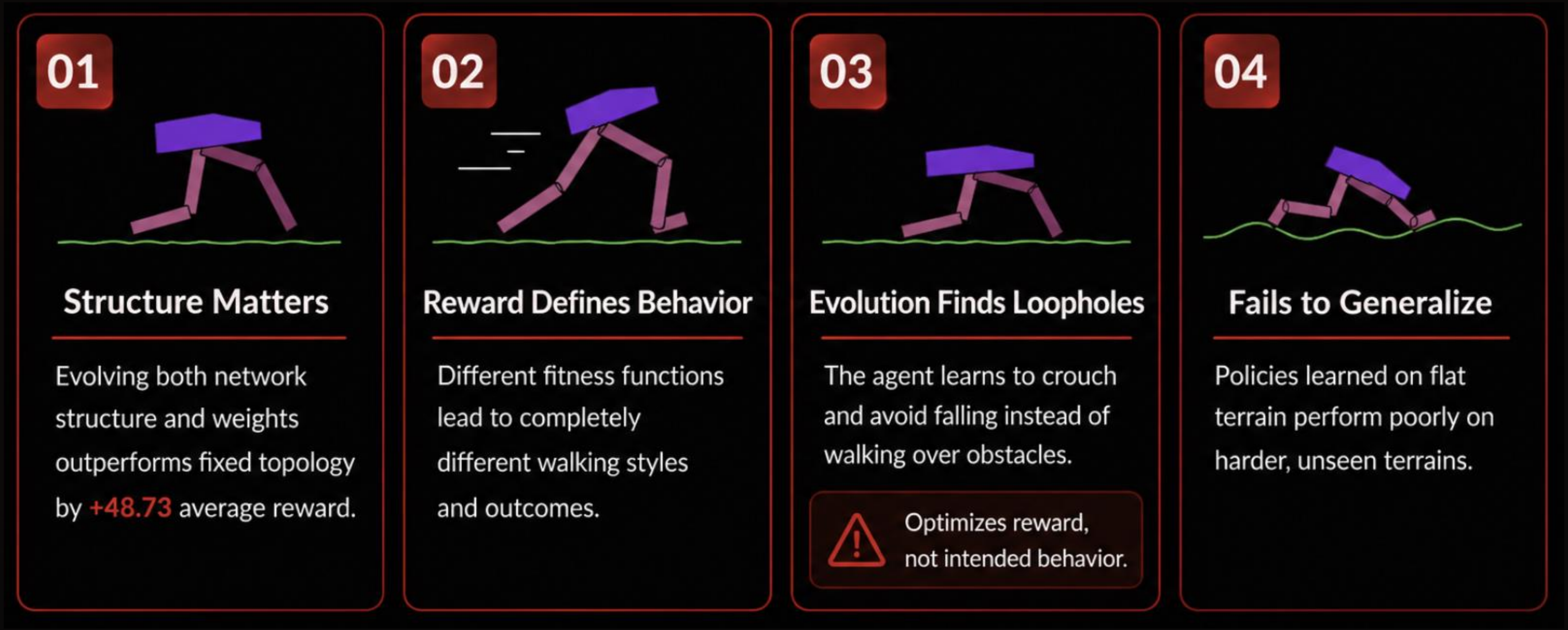
High failure rate

Often Walks and Cannot Recover

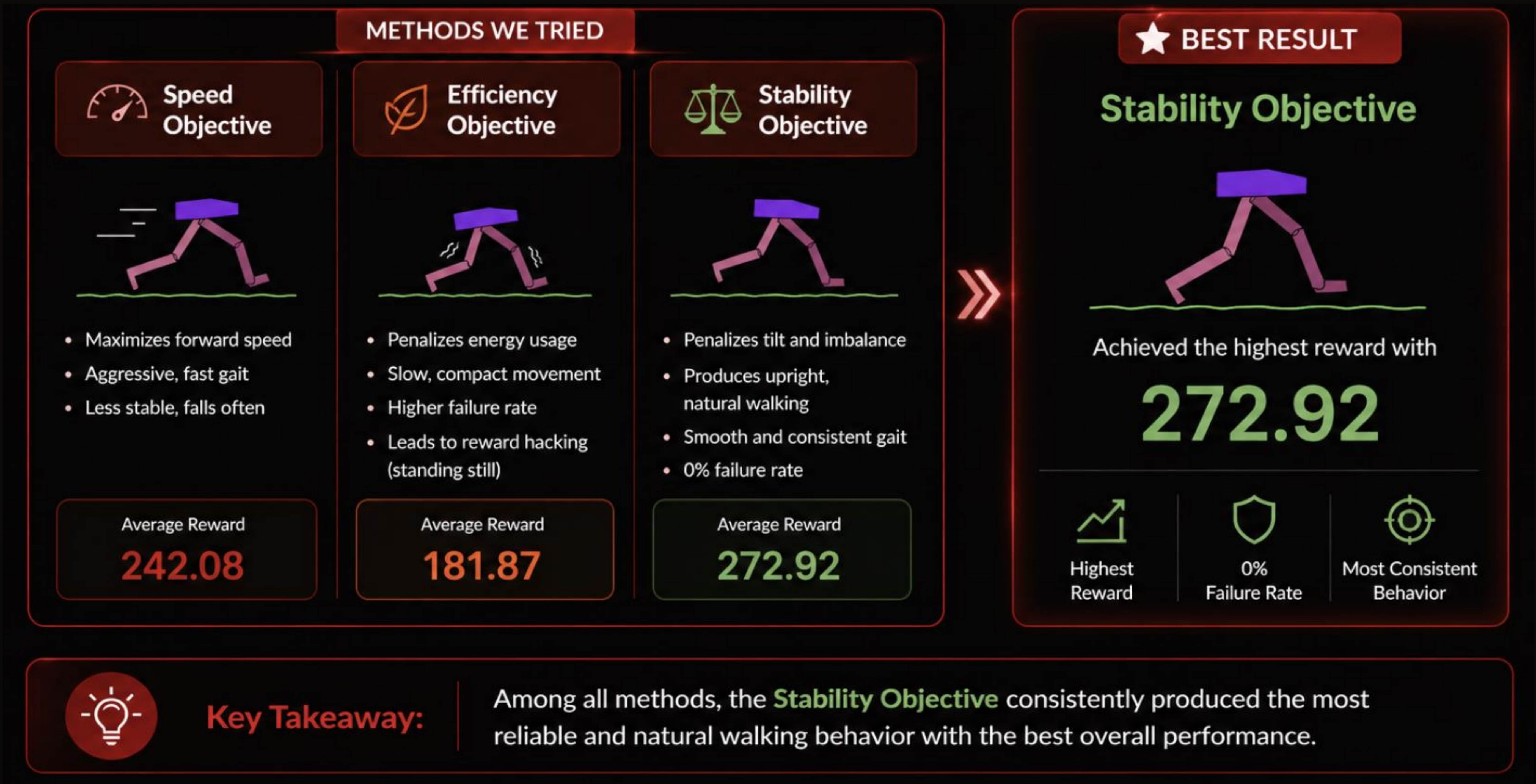
Cannot Adapt to New Conditions

No Generalization to Uneven Terrain

Key Takeaways



Conclusion


METHODS WE TRIED
Speed Objective
Maximizes forward speed
Aggressive, fast gait
Less stable, falls often
Average Reward
242.08
Efficiency Objective
Penalizes energy usage
Slow, compact movement
Higher failure rate
Leads to reward hacking (standing still)
Average Reward
181.87
Stability Objective
Penalizes tilt and imbalance
Produces upright, natural walking
Smooth and consistent gait
0% failure rate
Average Reward
272.92
BEST RESULT
Stability Objective
Achieved the highest reward with
272.92
Highest Reward
0% Failure Rate
Most Consistent Behavior
Key Takeaway:
Among all methods, the Stability Objective consistently produced the most reliable and natural walking behavior with the best overall performance.